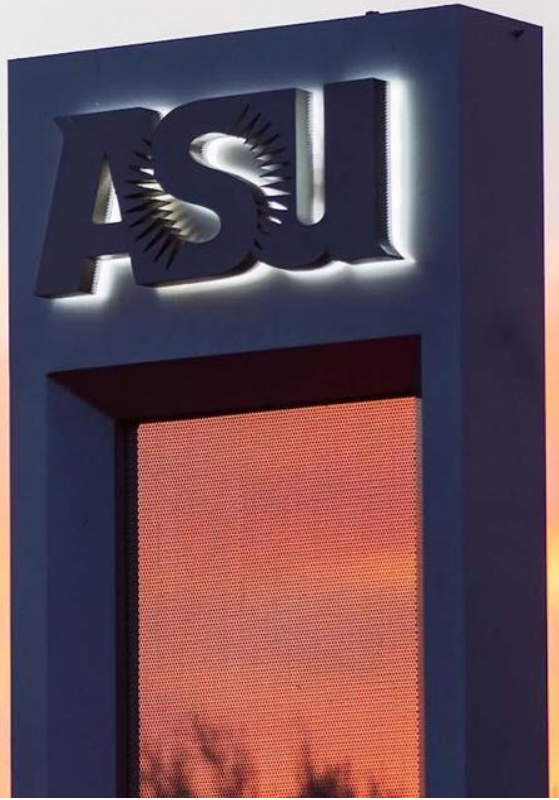
Wearable Biofeedback Device for Arm Swing



Arizona State University

Project 8 // Parkinson's Disease // SyncStride

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Agenda

1. Mission Statement
2. Client Need and Problem Definition
3. Background Information
4. Client Meeting Takeaways
5. Literature Review Brief
6. Ideation Findings
7. Concept Design

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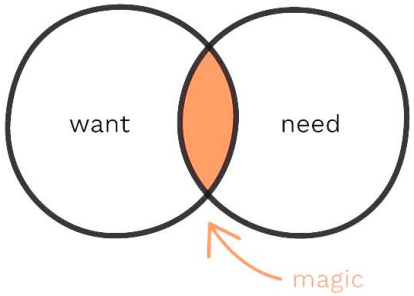
Mission Statement

United by a passion for solving real-world challenges through technology, we strive to bridge the gap between technology and human potential. Blending expertise in mechanical, electrical, mechatronics, and computer science, we will create innovative solutions that foster independence and mobility for all. We will design smart, human-centric solutions that improve healthcare and push the boundaries of technology.

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Client Need

Patients with Parkinson's Disease or similar conditions often experience reduced arm swing, affecting their gait, balance, and overall mobility. A wearable biofeedback device is needed to provide real-time vibrotactile or sound cues, helping them restore natural arm movement without requiring constant external assistance.



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Problem Definition

Many people with Parkinson’s struggle with reduced arm swing, affecting balance and movement. Existing solutions rely on external cues, which aren’t practical for independent walking. This project develops a wearable device that provides real-time feedback, helping users improve gait, restore movement, and prevent complications.

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Background Information

Parkinson’s Disease is a neurodegenerative disorder causing tremors, stiffness and impaired movement due to dopamine neuron loss. One of the symptoms is reduced arm swing while walking, which can lead to balance and gait issues. Therapists utilize a variety of techniques to help improve arm swing in people with Parkinson’s Disease, one of which is utilizing physical targets for the patients to try to reach while walking on a treadmill.



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Client Meeting Takeaways

Client Meeting Takeaways

1. Client requests a wearable assistive device for Parkinson's patients or patients suffering from some gait impairment. (arm movement under scope of study)
2. Client request's the deice to be designed initially for clinic use.
3. The target age group for the device is between 40-65 years.
4. Client insists on collecting some form of data for post processing and analysis.
5. Hit the target based approach is better than motion inducing approach for a gait scale between 0-2

Literature Review

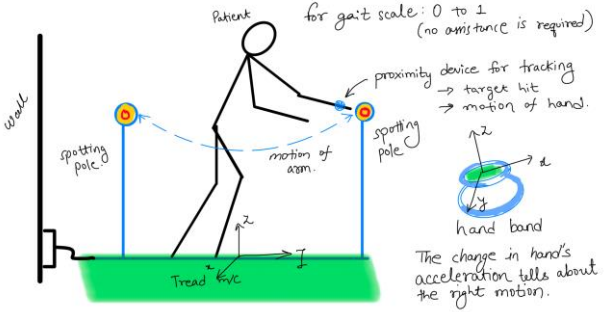
1. Parkinson's is a neurodegenerative disease and sensory disorder with no clear indication of it's cause of origin. Literature suggests problem with basal ganglia in brain.
2. Treatment is with dopamine, L-DOPA is being used to cross blood-brain barrier. Physical therapy is the best option with repetitive exercises.

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Ideation

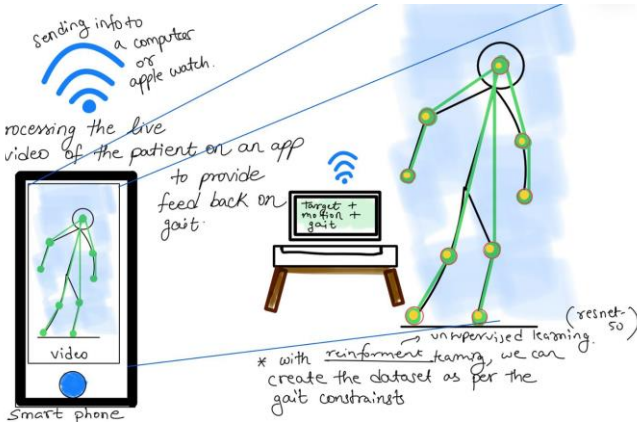
01

Wearable IMU device with sensory feedback (Light, Vibratory, Sound) to Validate Successful Motion



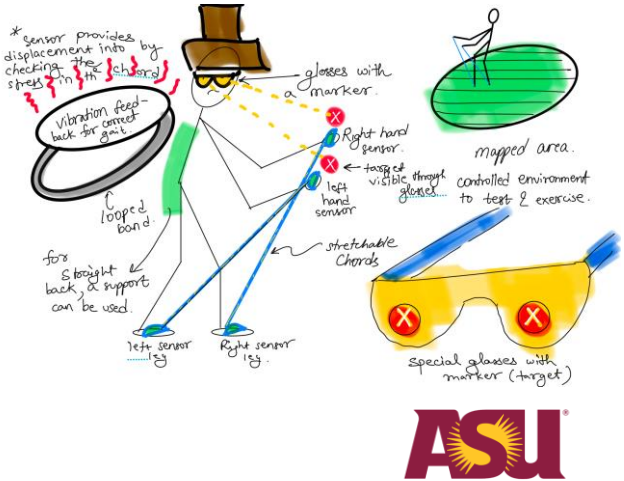
02

Motion monitor via video/AR



03

Resistance band/physically resistive material



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Ideation Continued

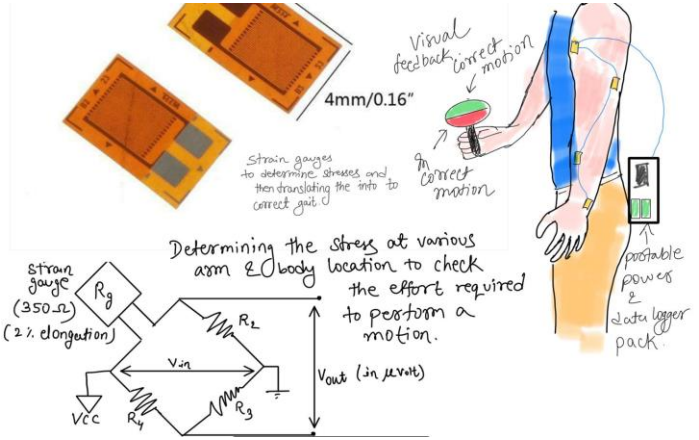
04

Creating an app for a pre-existing product (smart watch or nintendo joycons or a console)



05

Device utilizing analog sensors (potentiometer, flex sensor, strain gauge)



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Detailed Timeline

February

1. Component Selection
2. Part Ordering
3. Initial Design Review
   a. Software UML
   b. Detailed Design Sketch

March

1. Design Concept Development
2. Receive Parts
3. Development of Prototype
4. Design Review with Shannon (PT)

April

1. Final Physical Prototype
2. Testing and Validation

May

1. Presentation of Device

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